Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 23: Approved Solutions

Proof by Contrapositive

Covers Chartrand & Zhang, Section 3.3

Status of this set

Ungraded practice. Solutions released with it. The midterm is Wednesday, October 14 — this set is also good review of proof writing.

Part I: How to Practice This Material

Say which implication you are proving

The template

To prove $P \Rightarrow Q$ by contrapositive:

1. **Write down the contrapositive** $(\sim Q) \Rightarrow (\sim P)$ explicitly.
2. **Assume** $\sim Q$ and give a direct proof of $\sim P$.
3. **Conclude** by citing that an implication and its contrapositive are equivalent.

Step 1 is not optional. Writing it down is what keeps you from proving the converse by accident.

When to reach for it

Use the contrapositive when the **hypothesis is hard to use** and the **conclusion is easy to negate**. Typical trigger: the hypothesis involves n^2 or a product, while the conclusion is a simple parity statement.
Negating swaps them, and the easy one becomes your starting point.

Three mistakes to expect

1. **Proving the converse instead.** Negate *and* swap.
2. **Assuming $\sim P$ and deriving $\sim Q$.** That is the inverse, which is not equivalent.
3. **Not stating which implication is being proved.** Correct algebra on the wrong claim earns nothing.

Part II: Practice Problems

Section A — Write the contrapositive first

1. For each, write the contrapositive, then prove the original statement.

(a) Let $n \in \mathbb{Z}$. If $7n - 3$ is even, then n is odd.

(b) Let $n \in \mathbb{Z}$. If $n^2 - 6n + 5$ is odd, then n is even.

Approved solution

(a) *Contrapositive:* if n is even, then $7n - 3$ is odd.
Proof. Assume n is even, so $n = 2k$ for some integer k. Then

$$7n - 3 = 14k - 3 = 14k - 4 + 1 = 2(7k - 2) + 1.$$

Since $7k - 2$ is an integer, $7n - 3$ is odd. By Theorem 1.46 the original statement follows. ■
(b) *Contrapositive:* if n is odd, then $n^2 - 6n + 5$ is even.
Proof. Assume n is odd, so $n = 2k + 1$. Then

$$n^2 - 6n + 5 = (4k^2 + 4k + 1) - (12k + 6) + 5 = 4k^2 - 8k = 2(2k^2 - 4k).$$

Since $2k^2 - 4k$ is an integer, $n^2 - 6n + 5$ is even. ■

Note the -3 handled in (a). To reach $2b+1$ from $14k-3$, borrow: $14k-3 = 14k-4+1$. Writing $2(7k) - 3$ proves nothing, since -3 is not $+1$.

Section B — Choosing the method

2. For each statement, say whether a direct proof or a contrapositive proof is more practical, and why. Then prove it.

(a) If n is even, then $9n - 6$ is even.

(b) If n^3 is even, then n is even.

Approved solution

(a) Direct. The hypothesis "n is even" immediately gives $n = 2k$, which is usable.
Proof. Let $n = 2k$ for some integer k. Then $9n - 6 = 18k - 6 = 2(9k - 3)$, and $9k - 3$ is an integer, so $9n - 6$ is even. ■
(b) Contrapositive. The hypothesis "n^3 is even" gives $n^3 = 2k$, and taking a cube root produces nothing usable. The contrapositive's hypothesis is easy.
Contrapositive: if n is odd, then n^3 is odd.
Proof. Assume $n = 2k + 1$. Then

$$n^3 = 8k^3 + 12k^2 + 6k + 1 = 2(4k^3 + 6k^2 + 3k) + 1,$$

and $4k^3 + 6k^2 + 3k$ is an integer, so n^3 is odd. ■

The signal is in the hypothesis. In (a) it hands you an equation; in (b) it hands you a cube root. That is the whole basis for choosing.

Section C — Biconditionals

3. Prove: let $n \in \mathbb{Z}$. Then $5n - 2$ is odd if and only if n is odd.

Approved solution

Two implications are required.
($\Leftarrow$) **Direct.** Assume n is odd, so $n = 2k + 1$. Then

$$5n - 2 = 10k + 5 - 2 = 10k + 3 = 2(5k + 1) + 1,$$

and $5k + 1$ is an integer, so $5n - 2$ is odd.
($\Rightarrow$) **Contrapositive.** We show: if n is even, then $5n - 2$ is even. Assume $n = 2k$. Then $5n - 2 = 10k - 2 = 2(5k - 1)$, and $5k - 1$ is an integer, so $5n - 2$ is even.
Both directions hold, so the biconditional is proved. ■

The two halves used different methods, which is normal and often necessary. A biconditional is two independent obligations, and each may be met however is easiest.

Section D — Finding the flaw

4. A student offers this proof. Identify what is wrong.

Claim: let $n \in \mathbb{Z}$. If $3n + 4$ is odd, then n is odd.

Proof. Assume n is odd. Then $n = 2k + 1$, so $3n + 4 = 6k + 7 = 2(3k + 3) + 1$, which is odd. Hence the claim holds.

Approved solution

The student proved the converse. The claim is "if $3n + 4$ is odd, then n is odd," so a direct proof must *assume* $3n + 4$ *is odd*. The proof instead assumes n is odd and derives that $3n + 4$ is odd — which is the reverse implication.
The algebra is correct throughout. The proof is still worthless for the stated claim.
A correct proof, by contrapositive. We show: if n is even, then $3n + 4$ is even. Assume $n = 2k$. Then $3n + 4 = 6k + 4 = 2(3k + 2)$, and $3k + 2$ is an integer, so $3n + 4$ is even. ■

This is the error from Lecture 23's flawed proof, in a shorter costume. It is invisible if you only check the arithmetic, which is why the template insists on writing down which implication you are proving before you start.

Section E — Writing

5. In five to eight complete sentences, explain why proving the contrapositive proves the original statement, while proving the converse does not. Name the results from earlier in the term that settle each, and give a concrete pair illustrating the difference.

Approved solution

Model answer. An implication and its contrapositive have identical columns in a truth table, so they are logically equivalent — that is Theorem 1.46 — and establishing either one establishes the

other. A proof by contrapositive is therefore a proof of the original statement written in a more convenient form, not a substitute for it. The converse is different: $P \Rightarrow Q$ and $Q \Rightarrow P$ agree in only two of the four rows, so knowing one is true tells you nothing about the other. For a concrete pair, "if n is divisible by 4 then n is even" is true, and so is its contrapositive "if n is not even then n is not divisible by 4"; but its converse "if n is even then n is divisible by 4" is false, as $n = 6$ shows. So the contrapositive is a legitimate route to the same destination, while the converse is a different claim that merely reuses the same two pieces.

Full credit requires naming the equivalence and explaining why the converse is independent. A concrete counterexample is required for the second half.

Section F — Challenge (optional)

6. Prove: let $m, n \in \mathbb{Z}$. If mn is odd, then m and n are both odd.

Approved solution

Contrapositive: if m and n are *not* both odd, then mn is even.
"Not both odd" means at least one is even — note the De Morgan step. Without loss of generality assume m is even; the argument is identical if n is.
Proof. Assume m is even, so $m = 2a$ for some integer a. Then $mn = 2a \cdot n = 2(an)$, and an is an integer, so mn is even. ■

Two things worth noticing. Negating "both odd" correctly requires De Morgan from Week 1 — the negation is "at least one is even," not "both are even." And "without loss of generality" is legitimate here only because the two cases are genuinely symmetric; if they were not, both would have to be written out.

Instructor notes

- **Problem 4 is the item to grade hardest.** The algebra is flawless and the proof is wrong. Students who cannot see it will repeat the error on the midterm.
- **Problem 2** teaches method selection, which is the real skill of Section 3.3. The signal is always in the hypothesis.
- **Problem 3** shows the two halves of a biconditional using different methods. Students often assume both halves must match.
- **Problem 6's De Morgan step** is where most attempts break. "Not both odd" means at least one is even.
- **Problem 1(a)**: reaching $2b + 1$ from $14k - 3$ requires borrowing. Watch for $2(7k) - 3$.
- Reserved for Homework 9: Section 3.3 Exercises 2, 4, 6, 8, 10, 12 (even).